

Electricity Demand Forecasting Using AI

Project Lab Automation

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Created a dataset with following data

1. Power generation data
2. Power consumption
3. Day-ahead-price
4. Weather data
5. Temporal features

Open Ideas

1. Down sample to 1h-resolution
2. Include Corona period
3. Include school- and “semi” holidays

Problems

- Heat-electricity ratio fluctuates ($\approx 1 : 1.13$) and no reliable data exist
- No high-resolution demand data available specifically for Berlin
- Using the Energie Atlas and 50Hertz demand as a proxy for Berlin introduces inconsistencies

Solution

- Use the 50Hertz region directly and model its demand, since high-quality data are available

Changes

- Target variable is now electricity demand (grid load) instead of single-plant generation

Down sample to 1h-resolution: **implemented**

- Original resolution is hourly

Corona Period: **not implemented**

- Affected the data from 2020 to 2022
- Timeline of the pandemic based on “rbb” [4] and “Bundesgesundheitsministerium” [2]
- Introduced Intervention Severity Score
- No relevance for the model today

School holidays: **implemented**

- Scraped from schulferien.org [3]
- Public holidays checked across all federal states
- occurrences summed to create a weighted *School Holidays* variable

Very simple model used as a reference to assess whether more complex models are worth implementing

Implementation:

- *ForecasterEquivalentDate()* from Skforecast [1]
- Returns the value observed in the same period of the previous season
- e.g. the same working day from the previous week, the same hour from the previous day

Backtesting

Backtesting setup:

- Rolling-origin evaluation using TimeSeriesFold
- Forecast horizon: 24 hours
- Model is refit at each fold to simulate real-world operation
- Evaluation period: 2015–2024, hourly resolution

Metrics:

Metric	Description
MAE	Robust to outliers symmetric error measure
MSE	Penalizes large errors more strongly for detecting instability in forecasts
MAPE	Relative error measure comparable across different demand levels

Baseline Model – Backtesting

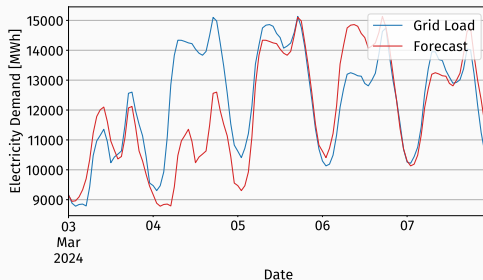


Figure 1: Seasonal Naive Forecaster.

MAE	MSE	MAPE
960.2 MWh	$1.93 \cdot 10^6 \text{ MWh}^2$	7.54%

Autoregressive Model

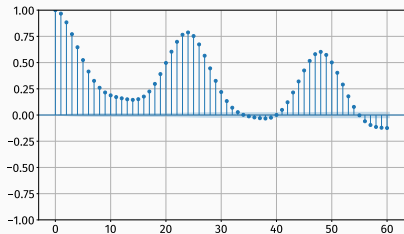


Figure 2: Autocorrelation

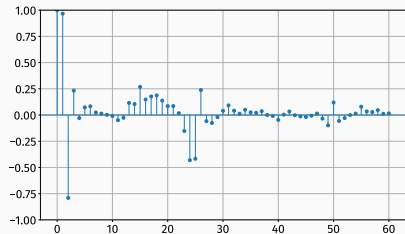


Figure 3: Partial Autocorrelation

Strong correlation between demand in one hour and prior hours and the corresponding hour in preceding days

Recursive Multi-Step Forecasting

Recursive autoregressive model *ForecasterRecursive*

Using different estimator to predict energy demand for the next 24 hours

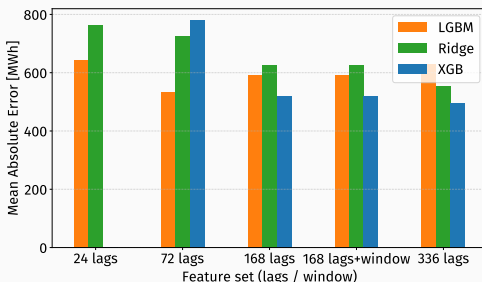


Figure 4: Model comparison with hyperparameters left to default.

Hyperparameter tuning

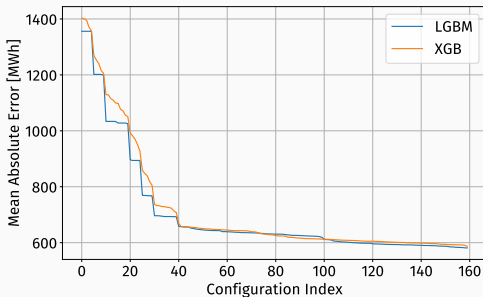


Figure 5: Hyperparameter Tuning using grid search.

Hyperparameter for XGBoost:

```
1 XGBRegressor(random_state=123, n_estimators=600,  
2                max_depth=8, learning_rate=0.05,  
3                gamma=5, min_child_weight=3,  
4                colsample_bytree=0.7, subsample=0.7)
```

Cyclic encoding of time-based features

- capture recurring seasonal patterns (daily, weekly, yearly)
- using trigonometric transformations:

$$x_{\sin} = \sin\left(2\pi \frac{t}{T}\right), \quad x_{\cos} = \cos\left(2\pi \frac{t}{T}\right) \quad (1)$$

- ensures a smooth transition between start and end of cycles

Exogenous Variables – Backtesting

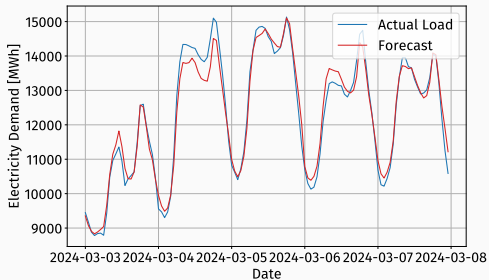


Figure 6: Forecast with LGBM-model and exogenous variables.

MAE	MSE	MAPE
238.4 MWh	$9.21 \cdot 10^5 \text{ MWh}^2$	1.92%

Road Map

1. Create a dataset 
2. Graphical exploration 
3. Comparing and choosing the best model 
4. Training and backtesting 
5. Exogenous Variables 
6. Feature selection (explanability and interpretability) 
7. Iterate through 3.-6. until satisfied with the results 
8. Deep Learning approach, Probabilistic forecasting 

References



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